

MIP_IMPACT_ANALYSIS_MODEL.md

Mainframe Intelligence Platform (MIP)

Enterprise Impact Analysis Architecture

Version: 1.0

Status: Core Intelligence Layer

Classification: Strategic Modernization Component

Owner: MIP Architecture Team

1. Executive Summary

Impact Analysis is one of the most valuable capabilities in MIP.

The Knowledge Graph tells us:

What is connected?

The Lineage Model tells us:

How things flow?

The Impact Analysis Model tells us:

What happens if we change something?

This capability enables organizations to safely modernize large-scale legacy systems while reducing operational and business risk.

2. Purpose

Impact Analysis determines the consequences of change.

Every modernization initiative eventually asks:

- What breaks if we modify this program?
- What downstream systems are affected?
- What business capabilities are impacted?
- What jobs will fail?

- What data flows will change?
- What is the modernization risk?

The Impact Analysis Engine provides evidence-backed answers.

3. Executive Vision

Without Impact Analysis:

Change Program

↓

Hope Nothing Breaks

With Impact Analysis:

Change Program

↓

Impact Analysis

↓

Affected Components

↓

Risk Assessment

↓

Mitigation Plan

↓

Safe Modernization

4. Why Impact Analysis Matters

Large enterprises often possess:

- 50,000+ Programs

- 10,000+ Batch Jobs
- Millions of Lines of Code
- Thousands of Tables
- Hundreds of Business Capabilities

A single change can trigger:

Program

↓

Tables

↓

Jobs

↓

Transactions

↓

Capabilities

↓

Customer Impact

Understanding these relationships is critical.

5. Core Objectives

The Impact Analysis Engine must answer:

Technical Impact

What programs are affected?

Data Impact

What tables are affected?

Operational Impact

What jobs are affected?

Business Impact

What capabilities are affected?

Modernization Impact

What migration plans are affected?

6. Impact Analysis Types

MIP supports six impact categories.

1. Program Impact
2. Job Impact
3. Data Impact
4. Transaction Impact
5. Capability Impact
6. Modernization Impact

7. Program Impact Analysis

Example Question

What happens if TS2AUTH changes?

Analysis

TS2AUTH

↓

Called By

↓

Calls

↓

Reads

↓

Writes

↓

Transactions

Outputs

Affected Programs

Affected Tables

Affected Transactions

Affected Jobs

Affected APIs

Affected Services

8. Job Impact Analysis

Example Question

What happens if JOB001 changes?

Analysis

JOB001

↓

Steps

↓

Programs

↓

Datasets

↓

Downstream Jobs

Outputs

Affected Batch Chains

Affected Data Flows

Affected Schedules

Affected SLAs

9. Data Impact Analysis

Example Question

What happens if CARD_MASTER changes?

Analysis

CARD_MASTER

↓

Readers

↓

Writers

↓

Consumers

↓

Capabilities

Outputs

Affected Programs

Affected Reports

Affected Services

Affected Capabilities

Affected Customers

10. Transaction Impact Analysis

Example Question

What happens if AUTHORIZATION changes?

Analysis

Transaction

↓

Programs

↓

Tables

↓

Capabilities

Outputs

Execution Paths

Dependencies

Business Risks

11. Capability Impact Analysis

Example Question

What happens if Card Activation changes?

Analysis

Capability

↓

Programs

↓

Jobs

↓

Tables

↓

Transactions

Outputs

Affected Systems

Affected Domains

Affected Customers

Affected Revenue Streams

12. Modernization Impact Analysis

Example Question

What happens if Authorization Service migrates?

Analysis

Migration Wave

↓

Capabilities

↓

Programs

↓

Data

↓

Dependencies

Outputs

Migration Risks

Wave Dependencies

Rollback Requirements

Mitigation Actions

13. Impact Severity Levels

Every impact receives a severity classification.

LOW

MEDIUM

HIGH

CRITICAL

14. Example Impact Assessment

Program:

TS2AUTH

Dependencies:

Calls: 83 Programs

Reads: 15 Tables

Writes: 12 Tables

Executed By: 14 Jobs

Supports: 3 Capabilities

Impact Rating:

CRITICAL

Reason:

High Dependency Concentration

Business Critical Function

Large Blast Radius

15. Impact Scoring Framework

Dependency Score

Measures:

Incoming Dependencies

Outgoing Dependencies

Shared Usage

Data Criticality Score

Measures:

Sensitive Data

Financial Data

Customer Data

Regulatory Data

Examples:

Cardholder Data

Payment Data

Personally Identifiable Information

Operational Score

Measures:

Batch Usage

Online Usage

Transaction Volume

Execution Frequency

Business Criticality Score

Measures:

Revenue Impact

Customer Impact

Compliance Impact

Operational Impact

16. Composite Impact Score

Formula:

Impact Score

=

Dependency Score

+

Data Criticality Score

+

Operational Score

+

Business Criticality Score

Range:

0-100

17. Impact Categories

Low Impact

Range:

0–25

Characteristics:

- Few dependencies
 - Limited usage
 - Easy rollback
-

Medium Impact

Range:

26–50

Characteristics:

- Moderate dependencies
 - Some operational impact
-

High Impact

Range:

51–75

Characteristics:

- Shared component
 - Significant business exposure
-

Critical Impact

Range:

76–100

Characteristics:

- Enterprise-wide dependency
- Major business risk
- Complex rollback requirements

18. Graph-Based Impact Analysis

Impact Analysis is powered by:

Metadata

+

Knowledge Graph

+

Lineage Graph

+

Reasoning Engine

19. Traversal Strategies

Upstream Impact

Question:

Who depends on this?

NetworkX:

```
nx.ancestors(graph, node)
```

Downstream Impact

Question:

What depends on this?

NetworkX:

```
nx.descendants(graph, node)
```

End-to-End Impact

Question:

```
Show complete blast radius.
```

NetworkX:

```
nx.all_simple_paths(graph, source, target)
```

20. Blast Radius Model

Represents total affected assets.

Example:

```
TS2AUTH
```

```
↓
```

```
83 Programs
```

```
↓
```

```
14 Jobs
```

```
↓
```

```
27 Tables
```

```
↓
```

```
3 Capabilities
```

```
↓
```

```
2 APIs
```

Blast Radius:

129 Assets

21. Risk Assessment Integration

Impact Analysis feeds directly into:

Risk Engine

Example:

High Impact
+
High Complexity
+
Poor Test Coverage
=
High Modernization Risk

22. Explainability Model

Every impact result must be explainable.

Example Output:

```
{
  "entity": "TS2AUTH",
  "impact_score": 89,
  "severity": "CRITICAL",
  "affected_assets": 129,
  "graph_paths": [],
  "evidence": [],
  "recommendations": []
}
```


23. Mitigation Recommendation Engine

For every HIGH or CRITICAL impact item, generate:

Testing Strategy

Rollback Strategy

Migration Strategy

Monitoring Strategy

Deployment Strategy

24. Leadership Dashboards

The platform should provide:

Critical Components Dashboard

Top 100 highest-risk components.

Blast Radius Dashboard

Components affecting the largest number of assets.

Modernization Risk Dashboard

High-risk modernization candidates.

Dependency Hotspot Dashboard

Programs with excessive coupling.

Migration Wave Dashboard

Wave readiness and migration status.

25. Enterprise Scale Targets

Support:

100,000+ Programs

20,000+ Jobs

500,000+ Tables

10,000,000+ Relationships

Millions of Lineage Paths

26. Integration Points

Consumes:

Metadata Repository

Knowledge Graph

Lineage Repository

Reasoning Engine

Produces:

Impact Reports

Risk Reports

Migration Recommendations

Executive Dashboards

27. Success Criteria

An architect should be able to ask:

What happens if I modify TS2AUTH?

And receive:

- Affected Programs
- Affected Jobs
- Affected Tables
- Affected Transactions
- Affected Capabilities
- Impact Score
- Blast Radius
- Risk Assessment
- Recommended Mitigations

within seconds.

28. Future Enhancements

Phase 2

Near Real-Time Impact Analysis

Phase 3

Predictive Risk Modeling

Phase 4

AI-Assisted Change Planning

Phase 5

Autonomous Modernization Recommendations

MIP Principle

Metadata identifies assets.

Knowledge Graph identifies relationships.

Lineage identifies movement.

Impact Analysis identifies consequences.

Reasoning identifies meaning.

Modernization Intelligence identifies opportunity.

End of Document